

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

	(Level/CO)	Marks
Q. 1 Solve Any Two of the following.		12
A) Describe mechatronics system with a neat sketch and give its practical examples.	(CO1)	6
B) Explain Force sensor (Load Cell) with a neat sketch by using strain gauge.	(CO1)	6
C) Discuss the working of AC Motor with a neat sketch and state its advantages.	(CO3)	6
Q.2 Solve Any Two of the following.		12
A) Describe the Data Acquisition System with a block diagram. List any two practical examples.	(CO2)	6
B) Draw a neat diagram of a Seven segment display system? Explain it with an example.	(CO2)	6
C) Explain with a neat sketch On/Off Controller.	(CO6)	6
Q. 3 Solve Any Two of the following.		12
A) Explain the sequencing of pneumatic cylinders using cascade method.	(CO3)	6
B) Give the comparison between pneumatic systems and hydraulic systems.	(CO3)	6
C) Derive model equation of a mechanical system for mass and spring.	(CO5)	6
Q.4 Solve Any Two of the following.		12
A) Explain with a block diagram 8081 microcontroller and its architecture.	(CO4)	6
B) Describe with a neat sketch Programmable Logic Controller (PLC).	(CO4)	6
C) Explain with a sketch of pressure sensor? Give few applications of it.	(CO1)	6
Q. 5 Solve Any Two of the following.		12
A) Explain with a neat sketch of Proportional- Integral -Derivative (PID) Controller.	(CO6)	6
B) Explain the Closed Loop Control System with a suitable example.	(CO5)	6
C) Describe with a neat sketch Proportional- Derivative (PD) Controller	(CO6)	6